INNOVATIVE TALENTS
SCIENCE TRAINING
PROGRAM

2024 SPRING QUARTER

Here is to certify that Jixi Gu has attended the course: Machine Learning and Data Analysis in the Innovative Talents Science Training Program from March 8th, to May 26th, 2024.

OFFICIAL TRANSCRIPT

Program	Professor	Class Period	Grade
Machine Learning and Data Analysis	Pradeep Ravikumar	03/08/2024~05/26/2024	A+

The curriculum design of the course focuses on drawing lessons from the educational concepts of both General Education and Research-Based Learning of world-class universities.

Professor: Pradeep Ravikumar
Home Institution: Carnegie Mellon University

GRADE	EQUIVALENT PERCENTAGES	GRADE	EQUIVALENT PERCENTAGES
A+	90-100	C+	67-69
A	85-89	C	63-66
A-	80-84	C-	60-62
B+	77-79	D+	57-59
B	73-76	D	53-56
B-	70-72	D-	50-52

Other Grading Information: Nonacademic Credit=Attended, Audited.

For more information visit ke.neoschool.com/student/#/passport/login and go to the Grading, Credits and Transcripts page in the Student Services Section.

Please note: The course syllabus and outlines are strictly in consistent with professor's home institutions. All lectures and readings are in English and all students works are also performed in English. Academy consistency is therefore maintained in accordance with the academic requirements at their respective colleges.

Neoscholar Education Group Inc. General Corporation SEAL 2020

ISSUED TO:

Name: Jixi Gu **Student ID:** KH10047946*

Class Period: 50 class hours

*Transcript valid only if bearing the Professor's Signature.

NEOSCHOLAR®
Educated by the 1%, Be the 1%

www.neoscholar.com

Machine Learning and Data Analysis

2024 .ITSTP

Basic Information

Course Title	Machine Learning and Data Analysis
Instructor	Pradeep Ravikumar, Associate Professor, Machine Learning Department, School of Computer Science, Carnegie Mellon University
Prerequisites	Proficiency in high-school level calculus, and linear algebra.
Required Text & Tools	Introduction to Data Mining, by P. Tan, M. Steinbach, V. Kumar, Addison Wesley, 2006. Programming Language: Python
Grading Criteria	Exam: 50% Homework: 40% Class Participation: 10%
Course Key Words	Data Analysis, Exploratory Data Analysis (EDA), Clustering, Classification, Anomaly Detection, Deep Learning

Schedule

No.	Topics
Lecture 1	Introduction to Data Analysis
Lecture 2	Exploratory Data Analysis (EDA), Clustering: Intro
Lecture 3	Clustering: Partitional, Hierarchical
Lecture 4	Classification: Decision Trees
Lecture 5	Classification: Nearest Neighbors, Ensembles
Lecture 6	Anomaly Detection, Syntax to Semantics, Boosting
Lecture 7	Regression Analysis and Predictive Modeling
Lecture 8	Feature Engineering and Selection
Lecture 9	Deep Learning
Lecture 10	Applied Machine Learning in Real-world Scenarios